Project - Retail Sales Data

1. **Project Overview :-**

This dataset captures retail transactions across multiple regions (East, West, North, South) with details on:

- Customers
- Product categories (Clothing, Electronics, Home, Sports)
- Order details (quantity, unit price, order date)
- Payment methods (Cash, Credit Card, UPI, Netbanking)

The goal is to answer client-style business questions such as revenue analysis, customer ranking, regional comparisons, and monthly sales trends.

2. **Data Structure :-**
   Columns :-

- OrderID → Unique identifier for each transaction
- CustomerID → Unique customer reference
- Region → Geographic region of sale
- ProductCategory → Category (Clothing, Electronics, Home, Sports)
- ProductName → Individual product identifier
- Quantity → Units sold
- UnitPrice → Price per unit
- OrderDate → Date of transaction
- PaymentMethod → Mode of payment

3. **Derived column to create :-**

- Revenue = Quantity × UnitPrice

- OrderValue = Revenue per transaction

4. **Key Insights :-**

- Electronics often have high unit prices (e.g., Product_6 at ₹815, Product_91 at ₹998).

- Clothing shows wide variation: some low-priced (₹120) and some premium (₹927).

- Sports products frequently have bulk quantities (7–9 units), driving higher order values.

- Payment methods are evenly spread, but Cash and Netbanking dominate large orders.

- ➢ Seasonal spikes: December shows heavy sales (holiday effect).
- ➢ Observed Patterns
- ➢ Regional differences :-
    - o East & West regions show strong Clothing and Electronics sales.
    - o South region has balanced sales across all categories.
    - o North region leans toward Home and Sports.
- ➢ Customer concentration: Some customers (e.g., IDs 1957, 1339, 1455) appear multiple times, suggesting repeat buyers.
- ➢ Monthly trends: Peaks around March, June, September, December → possible seasonal demand cycles.

❖ Client Questions :-

1. Which product category generated the highest revenue last quarter?

Solution :-

SQL

a. First add column Total plus order date in only month format
b. Use SUM(Total) grouped by ProductCategory.
c. Use where clause and between to find last 3 months revenue data
d. GROUP BY Region for average order value.
e. ORDER BY revenue descending for top customers.

Order_id	Customer	Region	Product_c	Product_	Quantity	Unit_price	Order_dat	Payment_method
1	1102	East	Clothing	Product_0	7	309	########	Credit Card
2	1435	West	Clothing	Product_1	4	405	########	Upi
3	1860	South	Home	Product_2	6	369	########	Netbanking
4	1270	West	Sports	Product_3	9	803	########	Cash
5	1106	South	Electronic	Product_4	1	786	########	Cash
6	1071	West	Home	Product_5	5	384	########	Upi
7	1700	East	Electronic	Product_6	7	815	########	Upi
8	1020	South	Clothing	Product_7	1	775	########	Netbanking
9	1614	East	Clothing	Product_8	4	862	########	Cash
10	1121	South	Sports	Product_9	6	326	########	Cash
11	1466	East	Clothing	Product_1	2	480	########	Cash
12	1214	East	Clothing	Product_1	5	559	########	Cash
13	1330	South	Home	Product_1	1	106	########	Cash
14	1458	South	Electronic	Product_1	6	128	########	Credit Card
15	1087	South	Electronic	Product_1	6	132	########	Upi
16	1372	North	Home	Product_1	4	745	########	Credit Card
17	1099	West	Sports	Product_1	8	978	########	Cash
18	1871	North	Electronic	Product_1	4	178	########	Netbanking
19	1663	North	Clothing	Product_1	8	558	########	Netbanking
20	1130	East	Electronic	Product_1	8	669	########	Cash
21	1661	South	Clothing	Product_2	5	637	########	Cash
22	1308	West	Sports	Product_2	4	581	########	Netbanking
23	1769	East	Home	Product_2	2	529	########	Credit Card
24	1343	North	Electronic	Product_2	3	837	########	Credit Card
25	1491	South	Home	Product_2	3	384	########	Cash
26	1413	East	Electronic	Product_2	4	615	########	Credit Card
27	1805	South	Clothing	Product_2	4	198	########	Netbanking
28	1385	East	Home	Product_2	5	496	########	Credit Card
29	1191	South	Electronic	Product_2	4	588	########	Cash
30	1955	North	Clothing	Product_2	8	138	########	Netbanking

SQL File 8* SQL File 7* SQL File 7* retail_sales

Limit to 1000 rows

```
SELECT * FROM df.retail_sales;
select product_category , sum(quantity*unit_price) as Total from retail_sales group by product_category order by total desc ;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

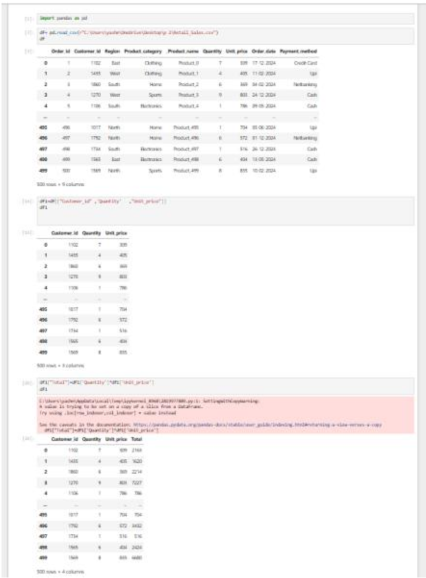
product_category	Total
Electronics	401103
Home	348000
Clothing	324255
Sports	305076

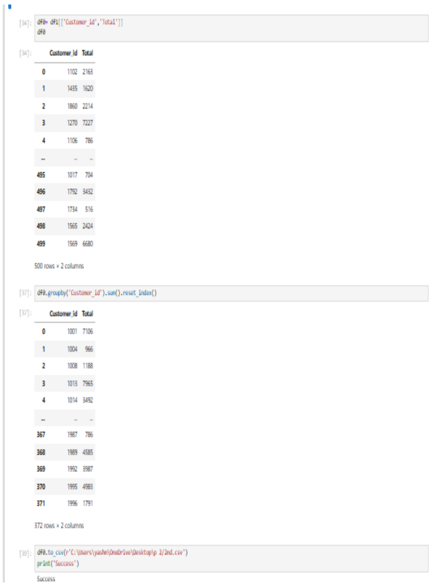
2. Identify the top 5 customers by total purchase value.

Solution :-

Python

a. First import csv file in python jupiter notebook with the help of pandas library .
b. Extract columns customer_id , quantity and unit_price .
c. Create new column in dataset with name Total .
d. Now last step do grouping by customer_id with total and export into power bi .





3. Compare average order value across different regions.

Solution :-

Excel

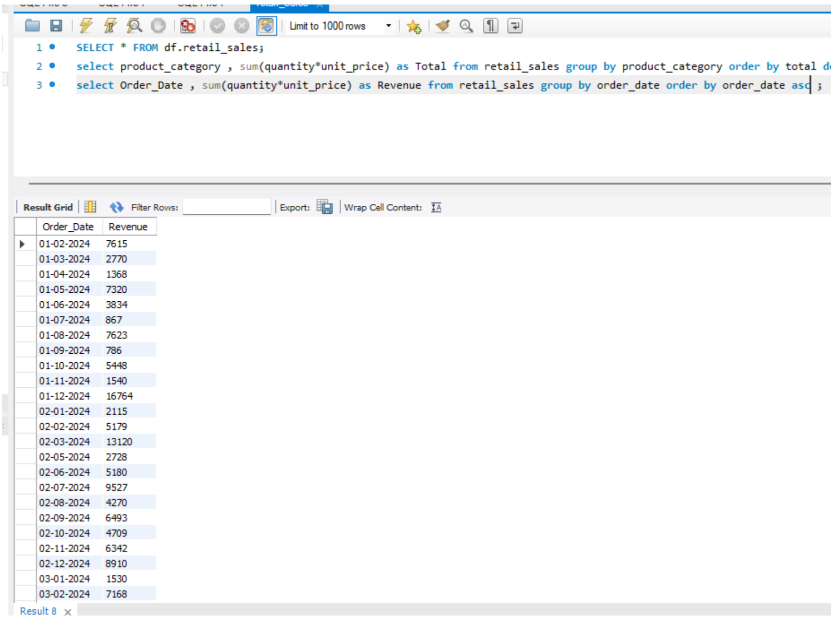
a. First open csv file in Excel .
b. Now create pivot table according to region wise average of quantity .

Row Labels	Average of Quantity
East	5.351851852
North	4.885714286
South	4.691666667
West	5.136363636
Grand Total	**5.006**

4. Find monthly sales trends for the last 12 months.
5. **Solution :-**

   My Sql

   a. Open my sql again and create a query with subquery
   b. Use group by function on order_date plus also aggregate on quantity * unit_price .
   c. Now export file in csv and load in power bi software .



```
1 ● SELECT * FROM df.retail_sales;
2 ● select product_category , sum(quantity*unit_price) as Total from retail_sales group by product_category order by total d
3 ● select Order_Date , sum(quantity*unit_price) as Revenue from retail_sales group by order_date order by order_date asc ;
```

Order_Date	Revenue
01-02-2024	7615
01-03-2024	2770
01-04-2024	1368
01-05-2024	7320
01-06-2024	3834
01-07-2024	867
01-08-2024	7623
01-09-2024	786
01-10-2024	5448
01-11-2024	1540
01-12-2024	16764
02-01-2024	2115
02-02-2024	5179
02-03-2024	13120
02-05-2024	2728
02-06-2024	5180
02-07-2024	9527
02-08-2024	4270
02-09-2024	6493
02-10-2024	4709
02-11-2024	6342
02-12-2024	8910
03-01-2024	1530
03-02-2024	7168

POWER BI REPORT :-